

Recap on Convolutional Neural Networks

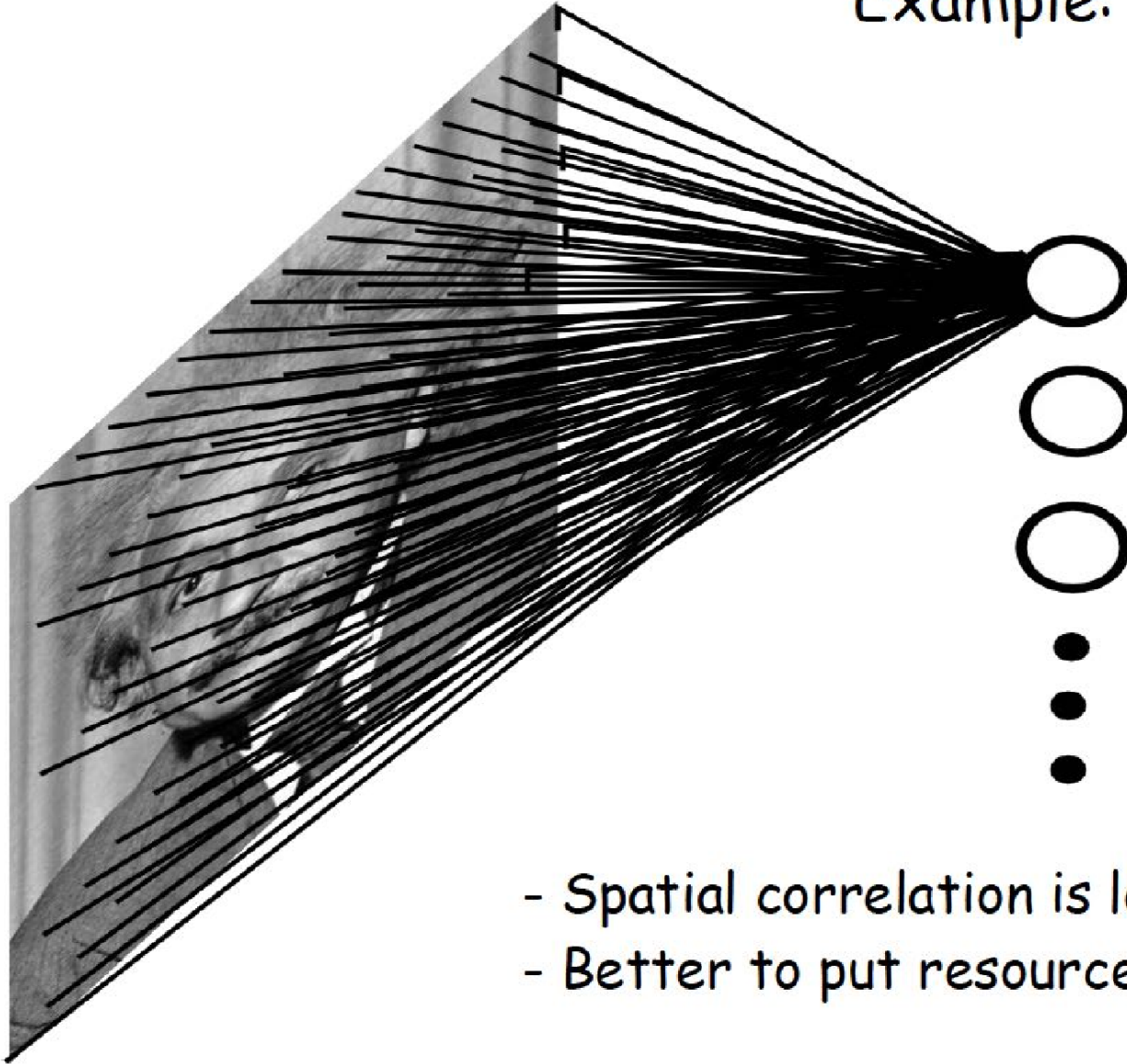
Advanced Machine Learning and Computer Vision
28 September 2026
Prof. Fabio Galasso



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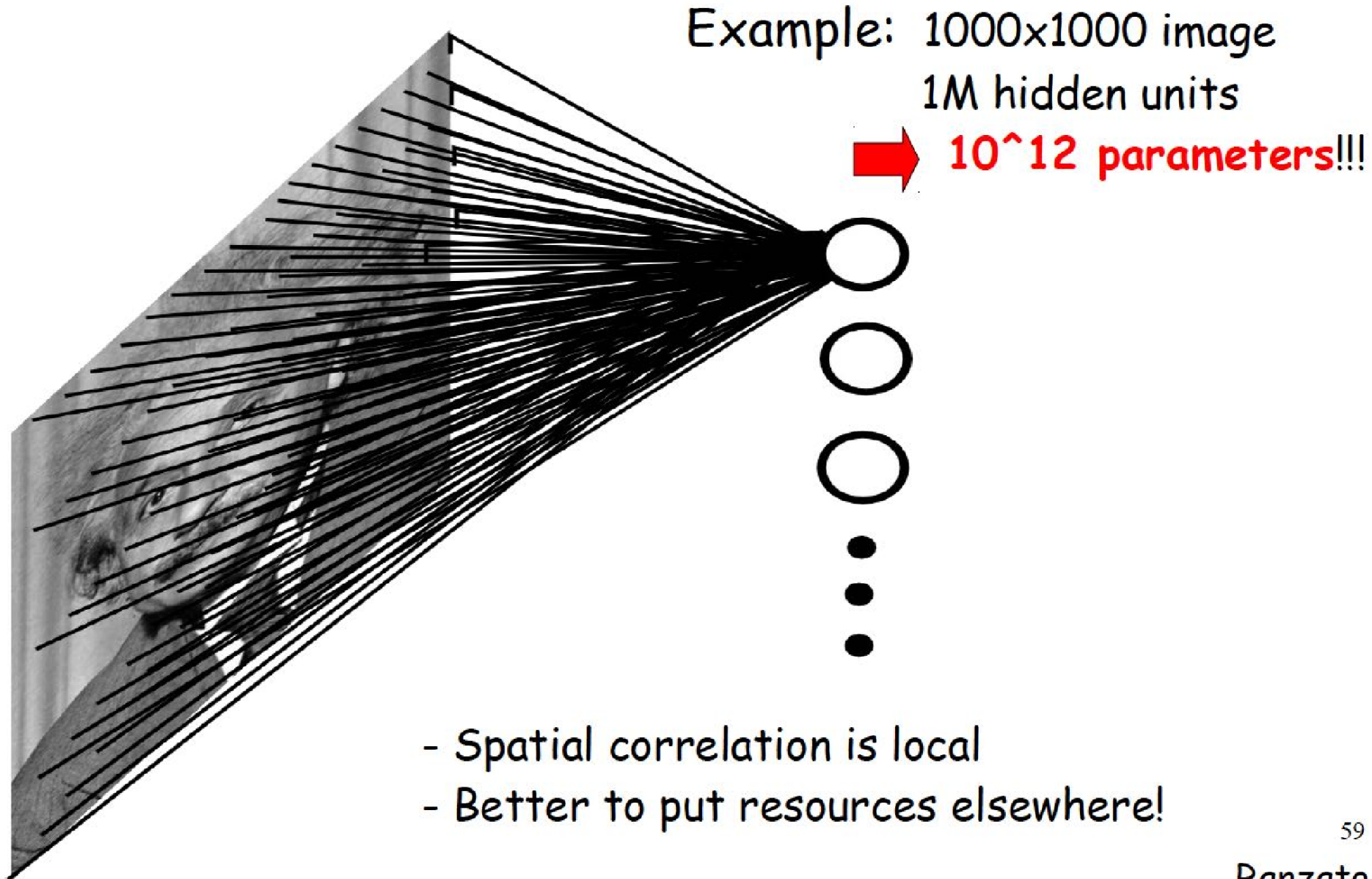
FULLY CONNECTED NEURAL NET

Example: 1000x1000 image
1M hidden units

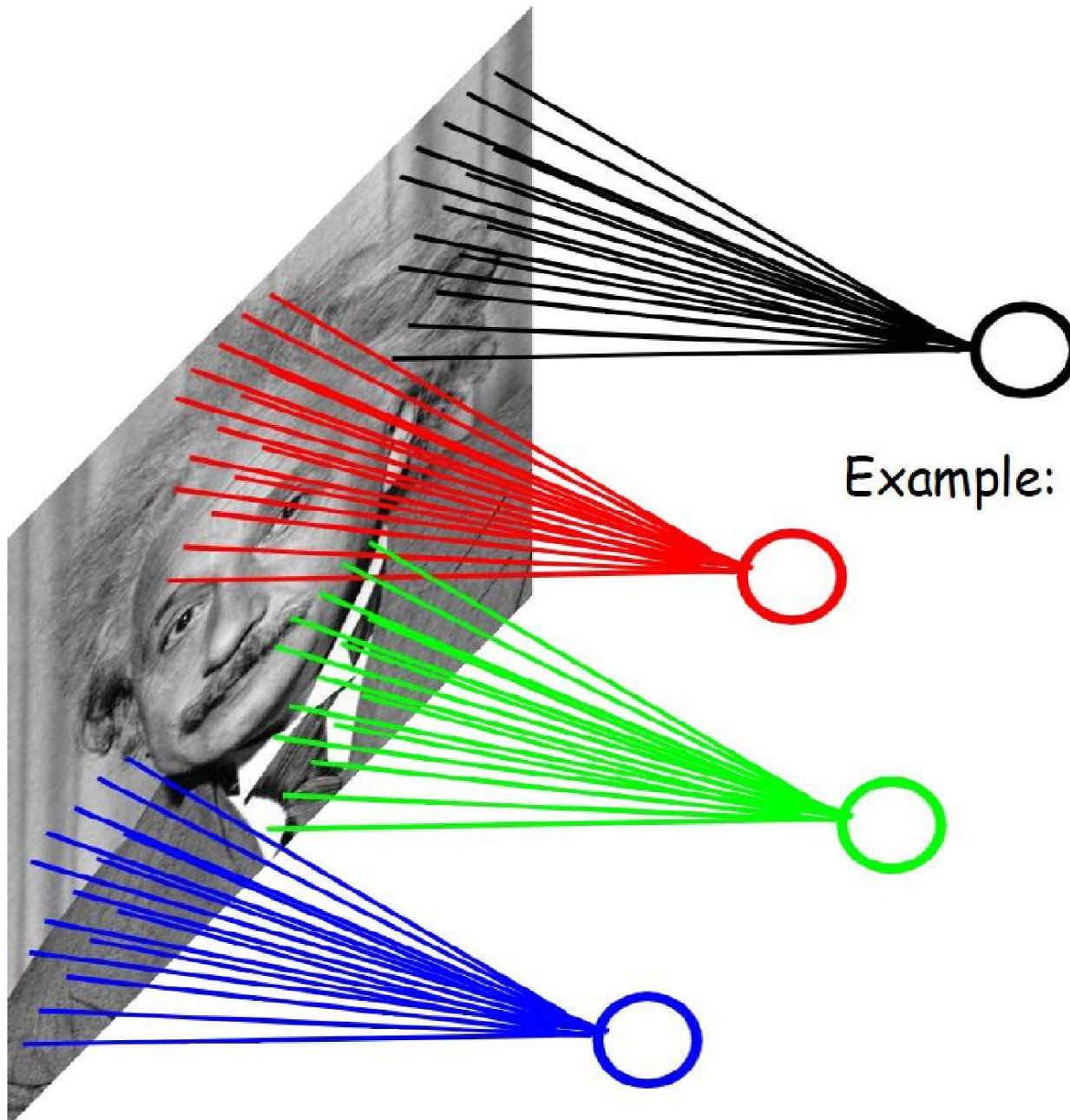


- Spatial correlation is local
- Better to put resources elsewhere!

FULLY CONNECTED NEURAL NET

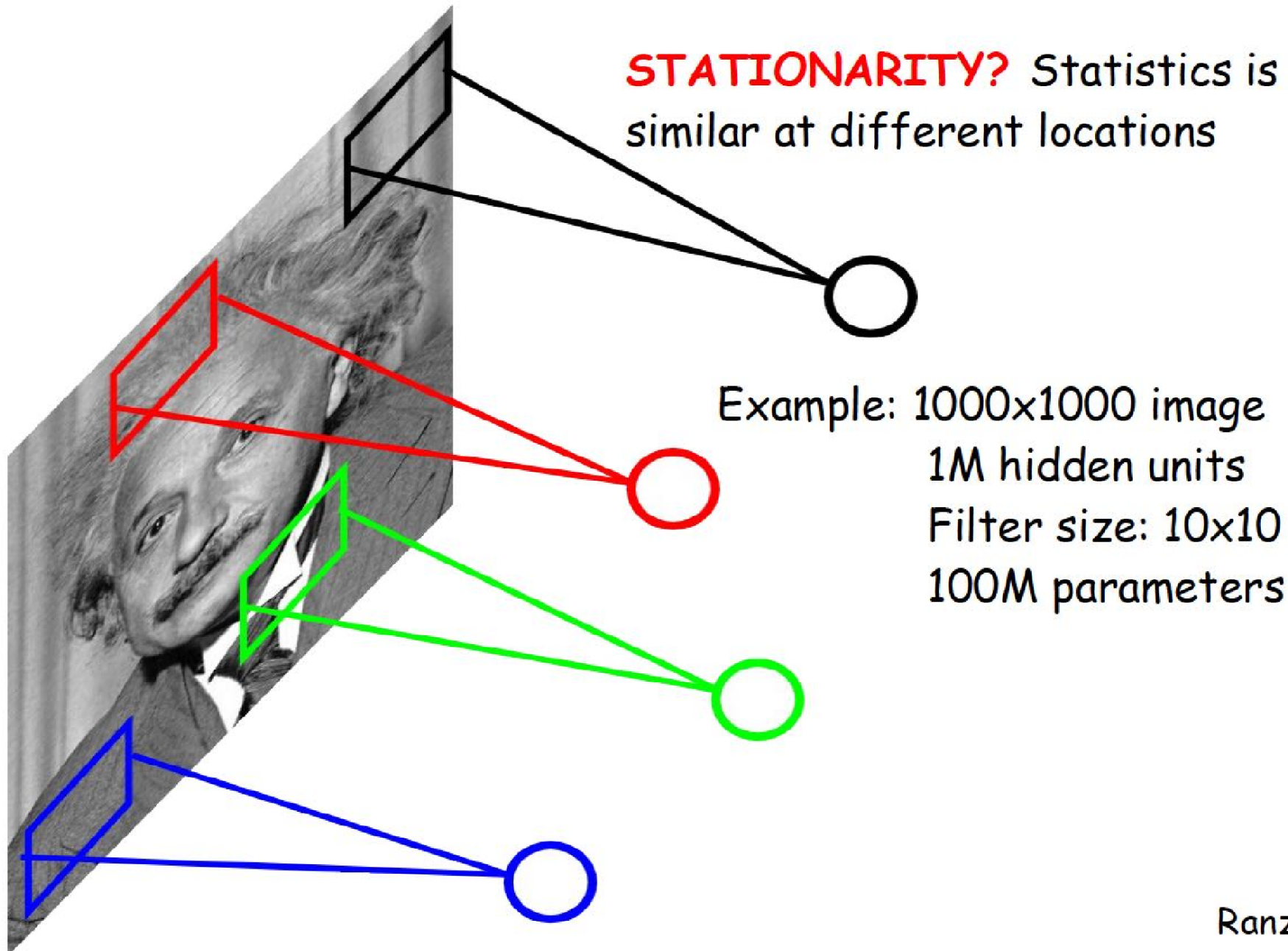


LOCALLY CONNECTED NEURAL NET

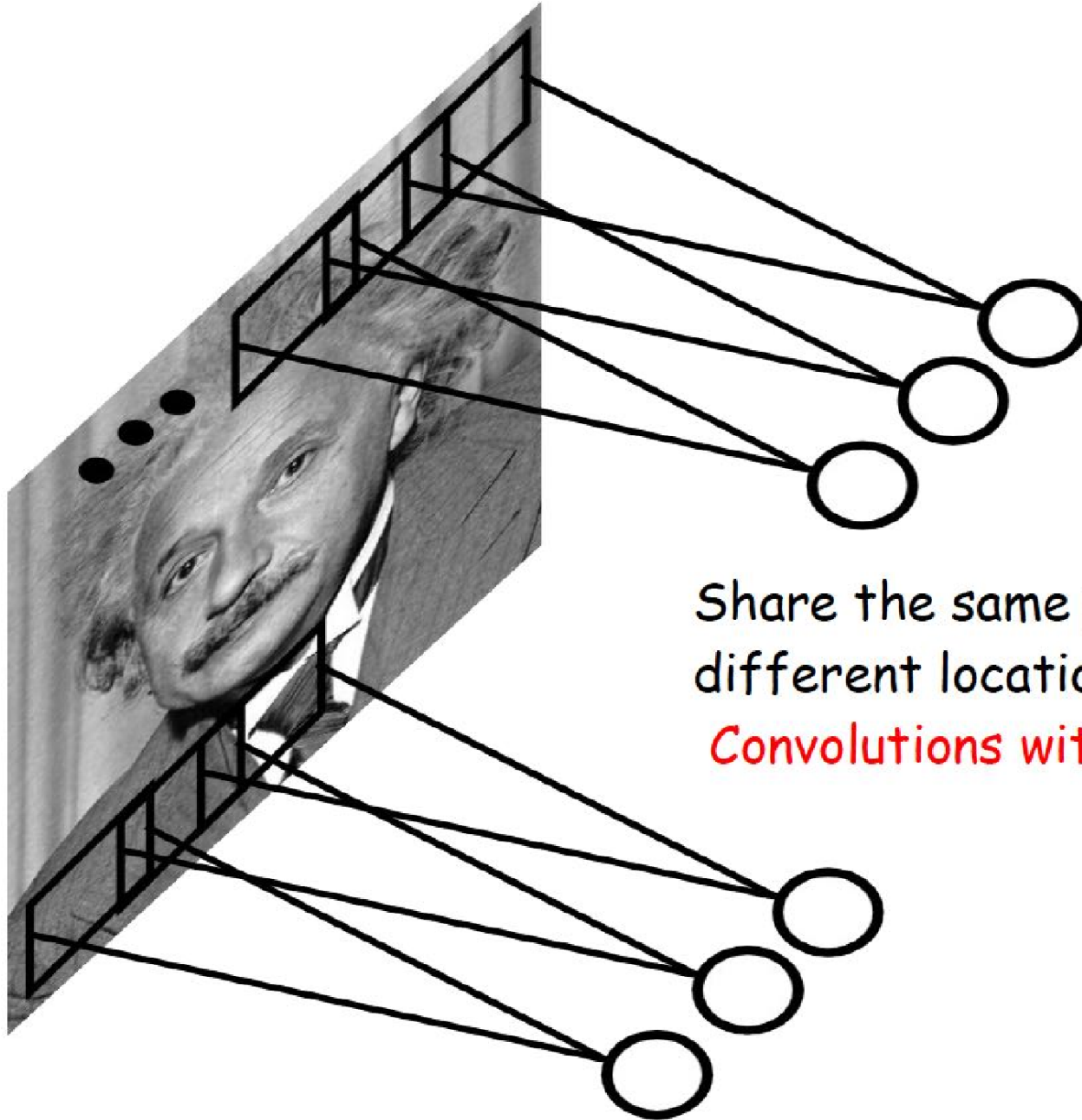


Example: 1000x1000 image
1M hidden units
Filter size: 10x10
100M parameters

LOCALLY CONNECTED NEURAL NET



CONVOLUTIONAL NET



Share the same parameters across
different locations:

Convolutions with learned kernels

CONVOLUTIONAL NET

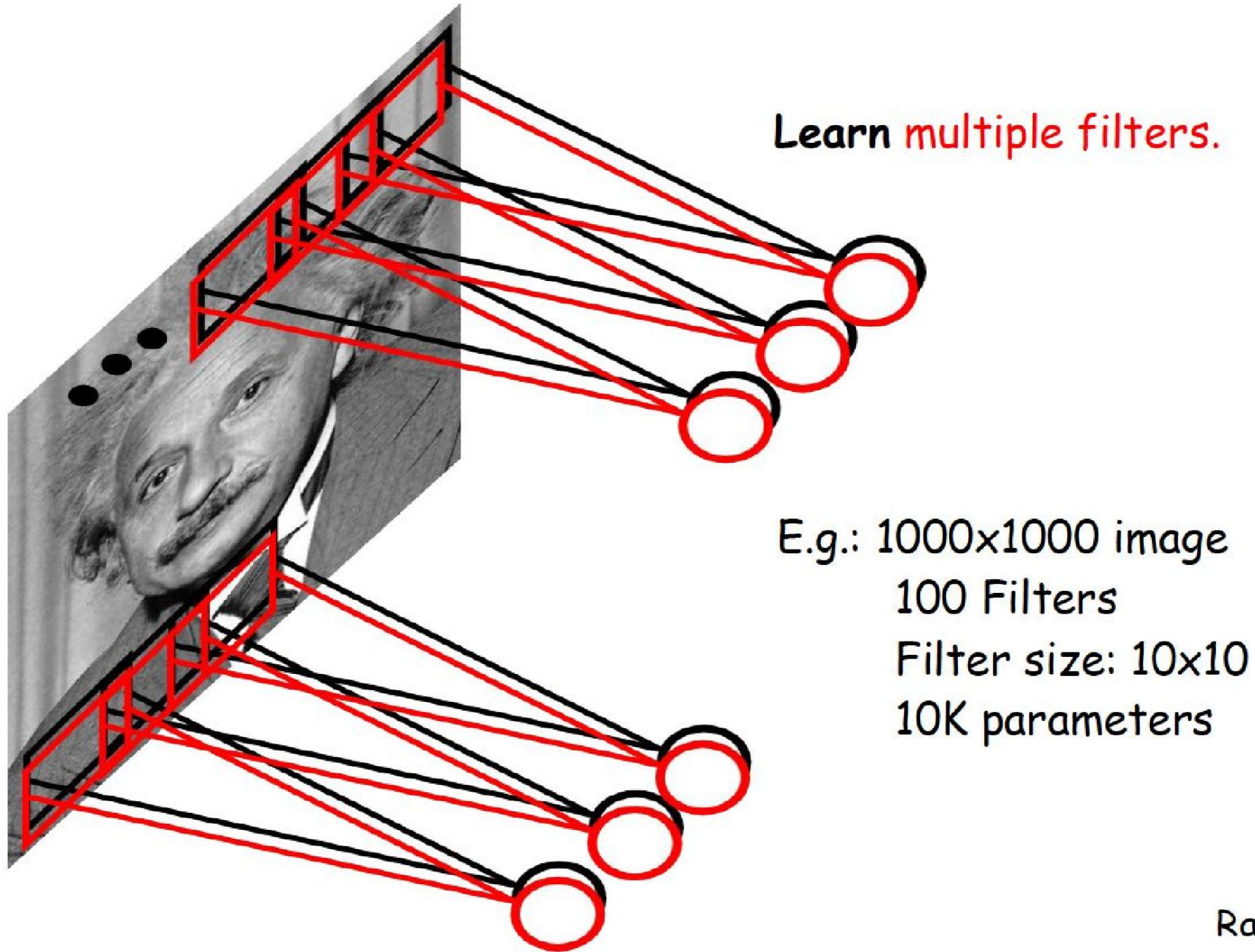


Image Filtering: 2D Convolution

- Convolution is linear filtering:
 - replace each pixel by a linear combination of its neighbors

$$f[m, n] = I \otimes g = \sum_{k, l} I[m - k, n - l] g[k, l]$$

- 2D convolution (discrete):
 - discrete Image: $I[m, n]$
 - filter 'kernel': $g[k, l]$
 - 'filtered' image: $f[m, n]$

$$f[m, n] = I[k, l] \otimes g[k, l]$$

	18	

8	5	2
7	5	3
9	4	1

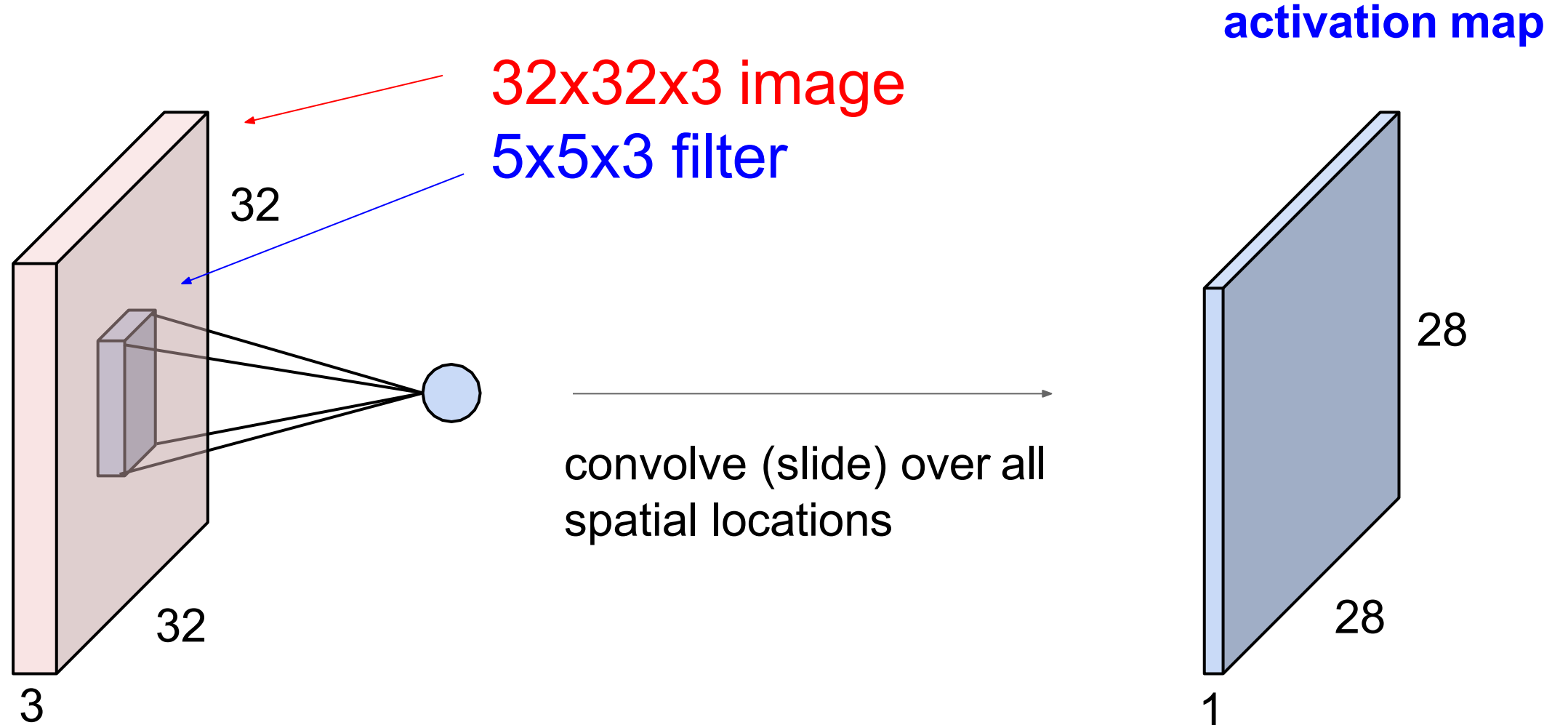
 $\otimes$

-1	0	1
-1	0	1
-1	0	1

Linear Systems

- Basic Properties:
 - homogeneity $T[a X] = a T[X]$
 - additivity $T[X_1 + X_2] = T[X_1] + T[X_2]$
 - superposition $T[aX_1 + bX_2] = a T[X_1] + b T[X_2]$
 - linear systems $\Leftrightarrow$ superposition
- examples:
 - matrix operations (additions, multiplication)
 - convolutions

Convolution Layer

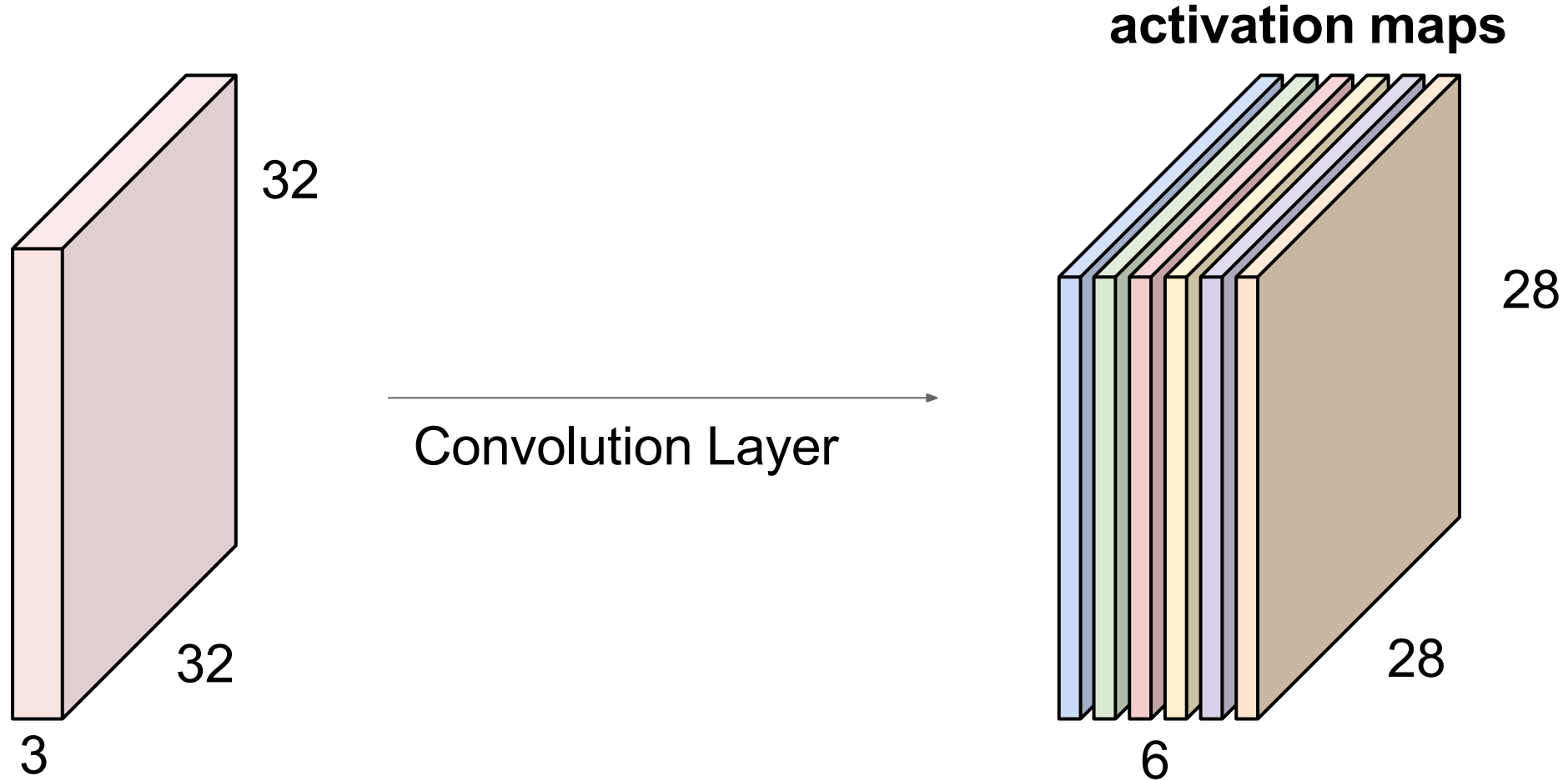


Convolution Layer

consider a second, **green** filter



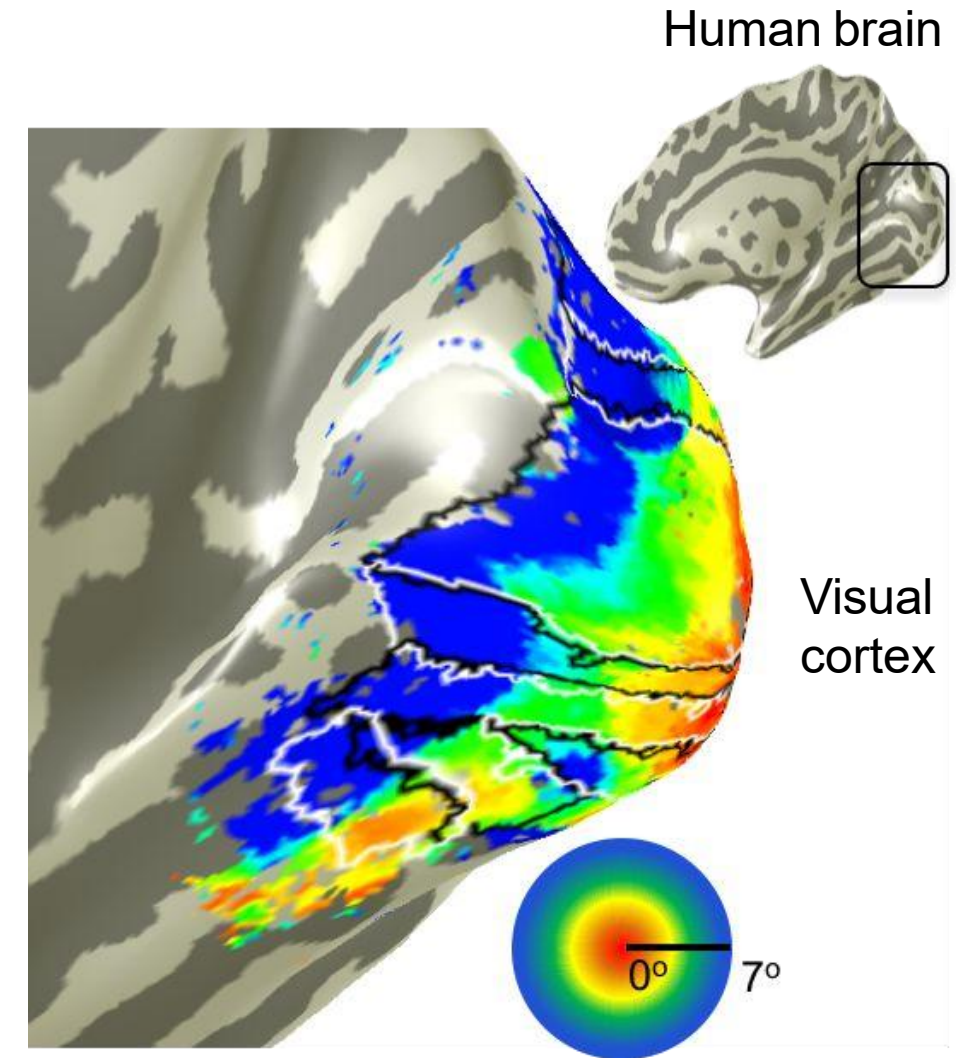
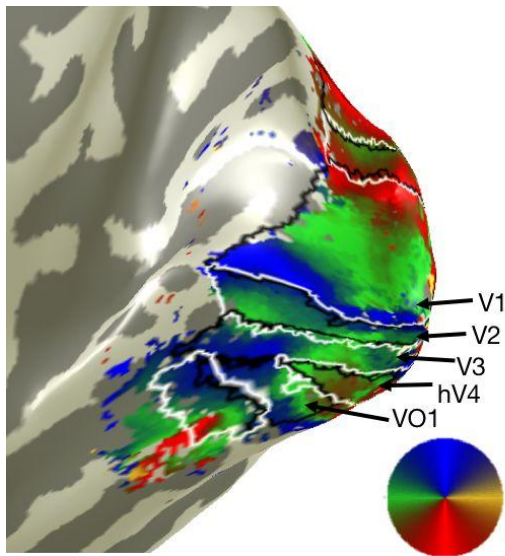
For example, if we had 6 5x5 filters, we'll get 6 separate activation maps:



We stack these up to get a “new image” of size 28x28x6!

A bit of history

Topographical mapping in the cortex:
nearby cells in cortex represent
nearby regions in the visual field

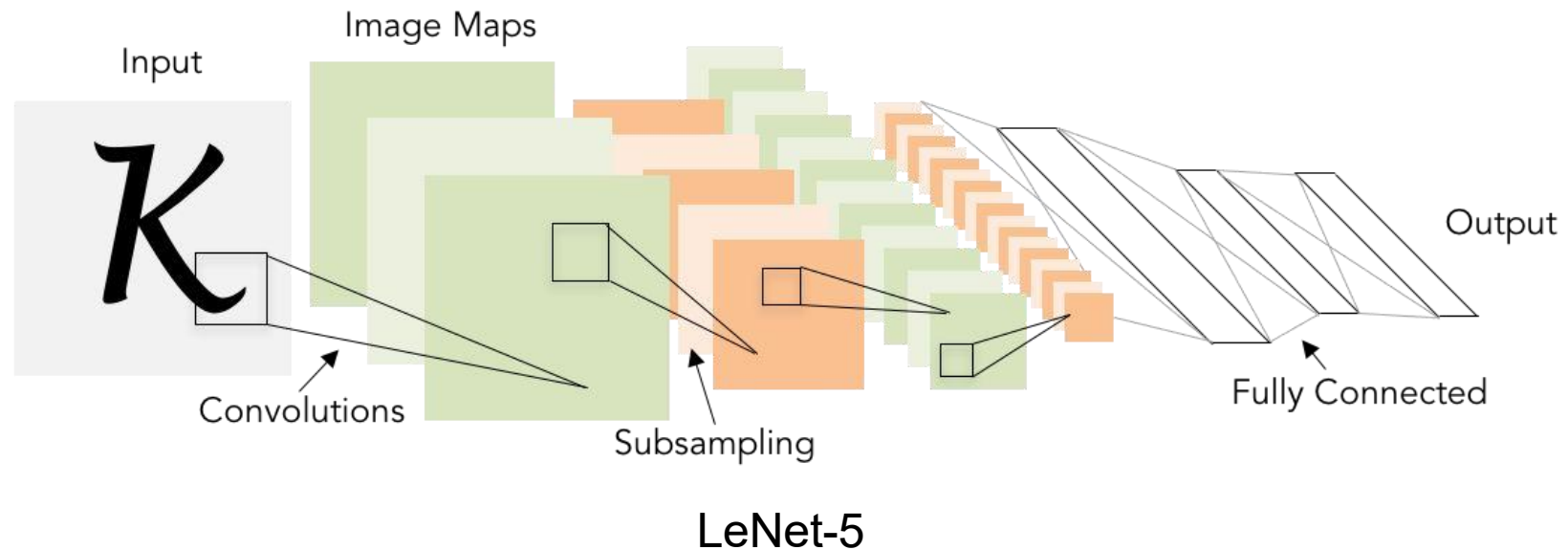


Retinotopy images courtesy of Jesse Gomez in the
Stanford Vision & Perception Neuroscience Lab.

A bit of history:

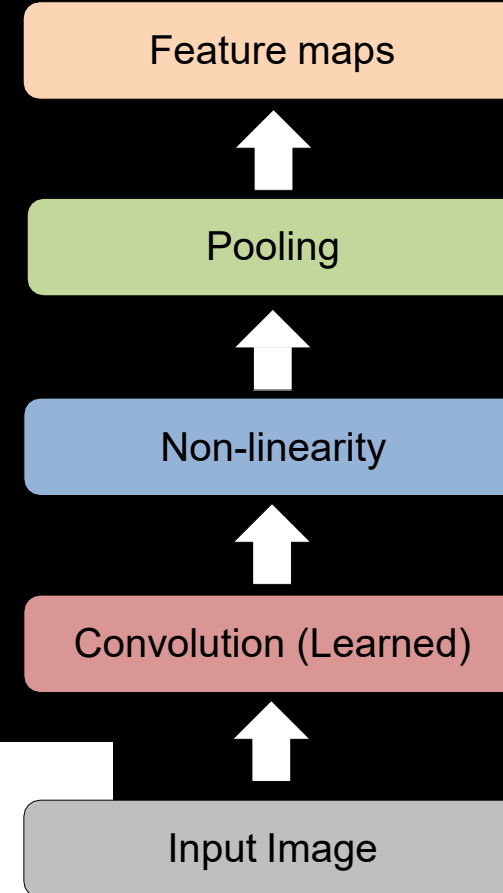
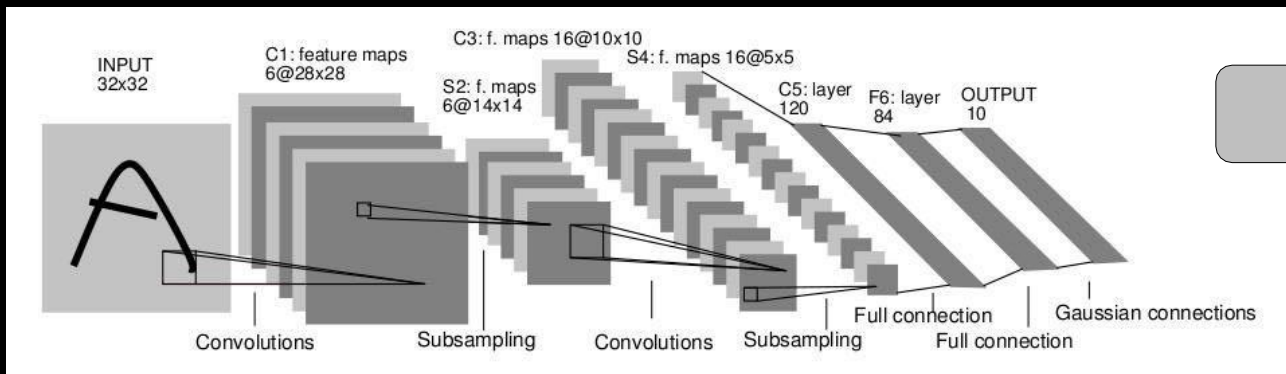
Gradient-based learning applied to document recognition

[LeCun, Bottou, Bengio, Haffner 1998]



Characteristics of LeNet

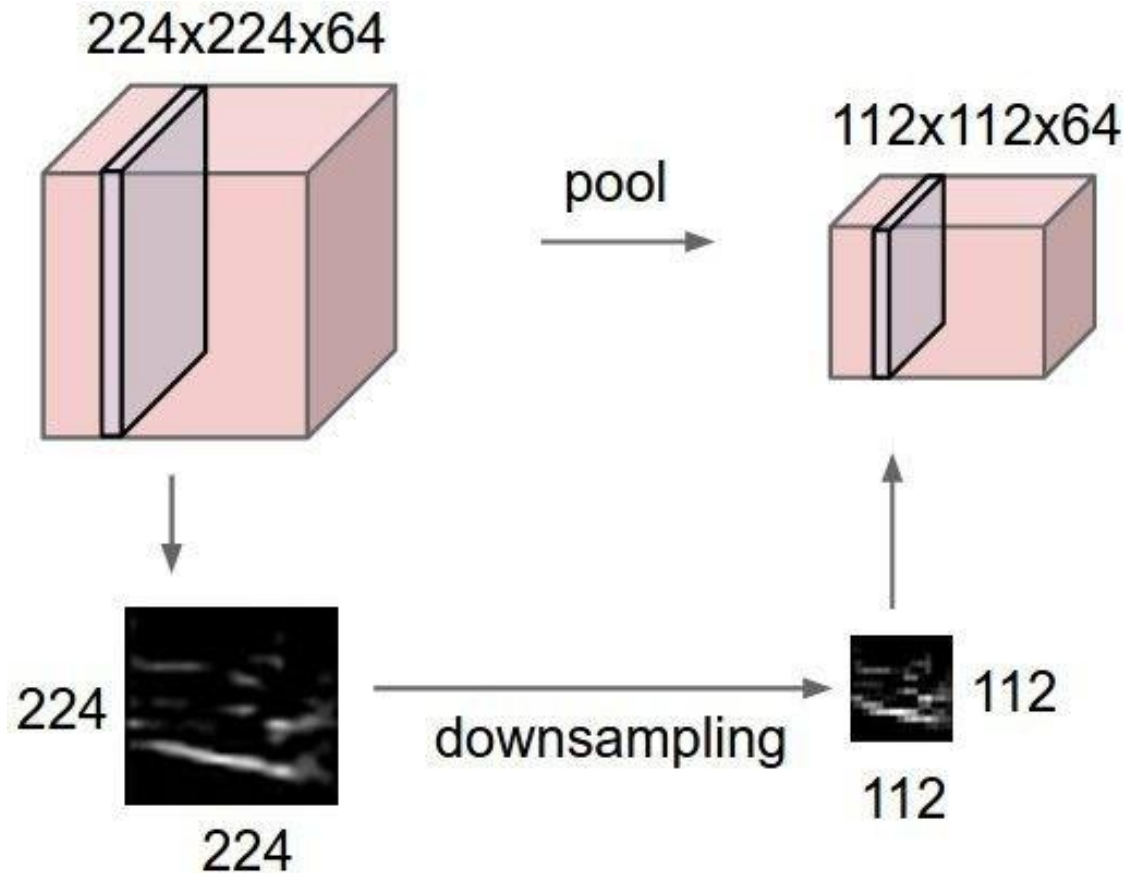
- Feed-forward:
 - Convolve input
 - Non-linearity (rectified linear)
 - Pooling (local max) / (=subsampling)
- Supervised
- Train convolutional filters by back-propagating classification error



[LeCun et al. 1989]

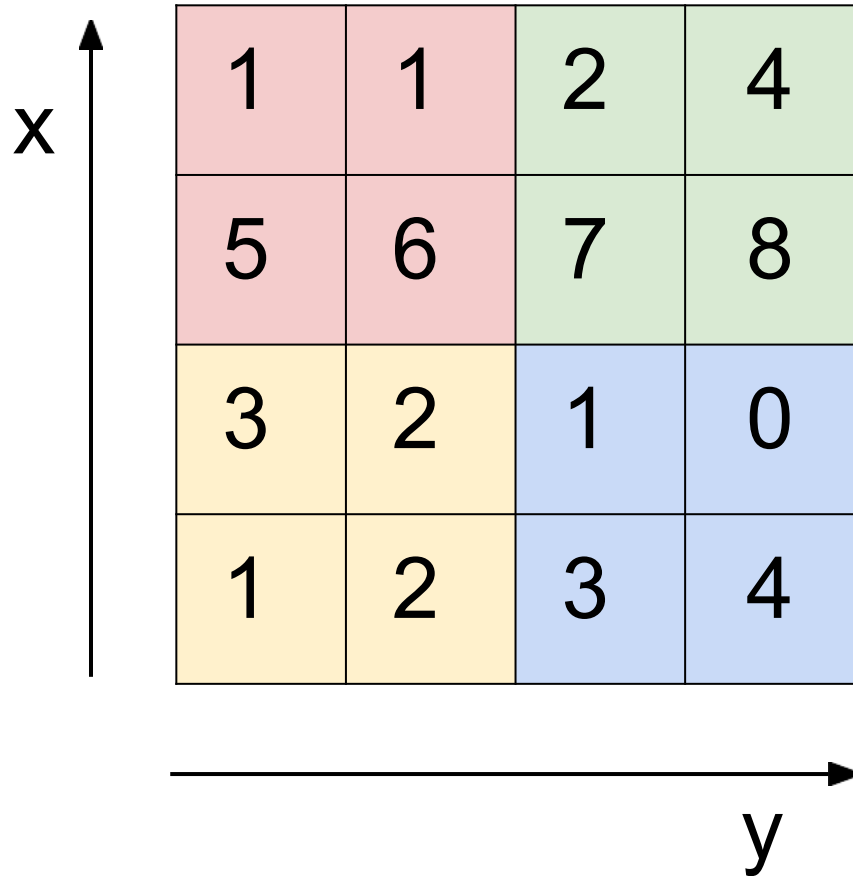
Pooling layer

- makes the representations smaller and more manageable
- operates over each activation map independently:

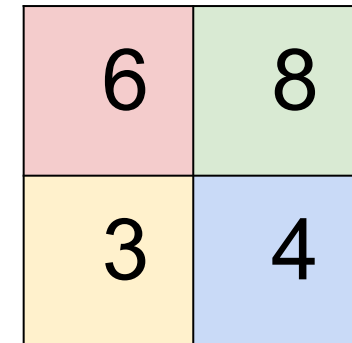


MAX POOLING

Single depth slice



max pool with 2x2 filters
and stride 2



Krizhevsky et al. [NIPS2012]

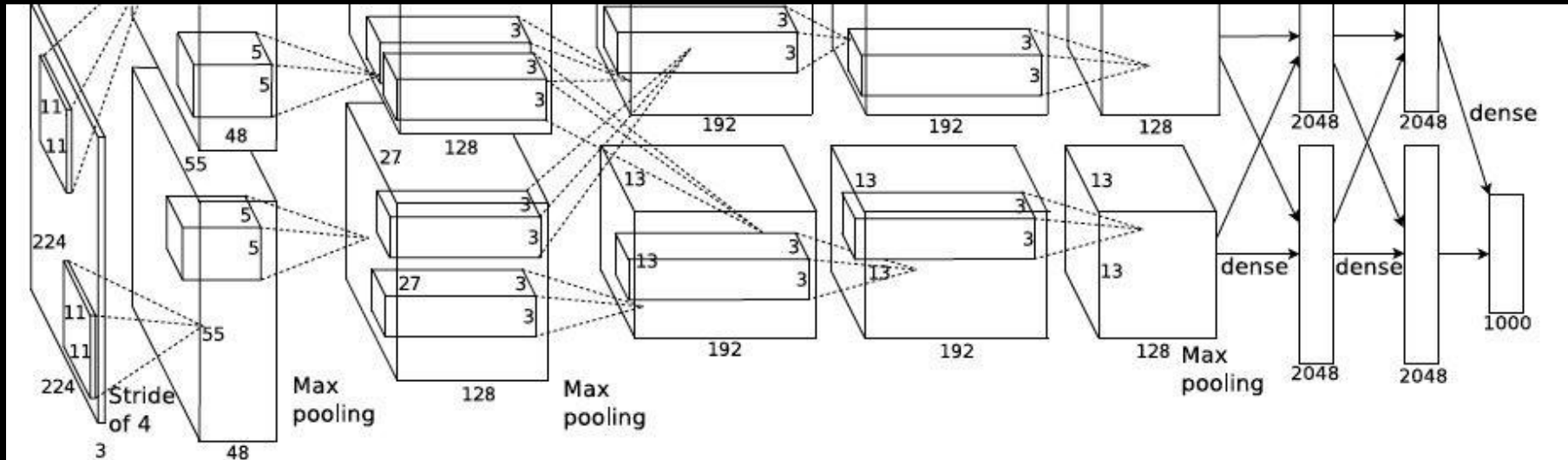
- Compared to LeCun'98 but:
 - Bigger model (8 layers)
 - More data (10^6 vs 10^3 images)
 - GPU implementation (50x speedup over CPU)
 - Better regularization (DropOut)

ImageNet Classification with Deep Convolutional Neural Networks

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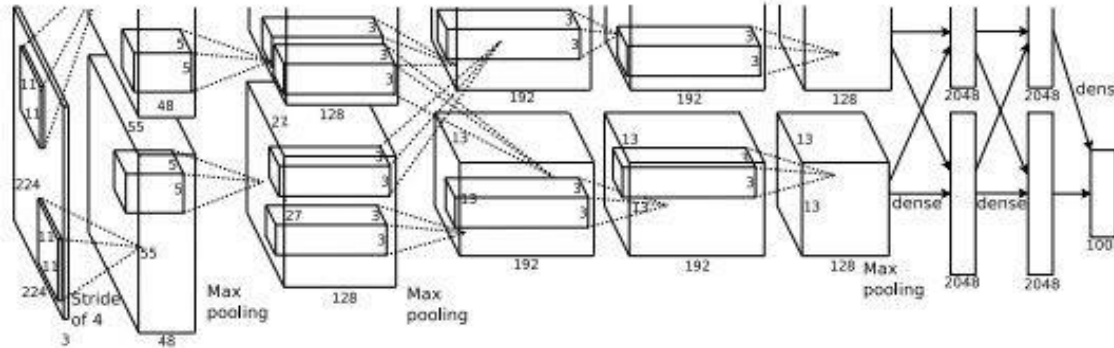


- 7 hidden layers, 650,000 neurons, 60,000,000 parameters
- Trained on 2 GPUs for a week

AlexNet

Imagenet classification with deep convolutional neural networks

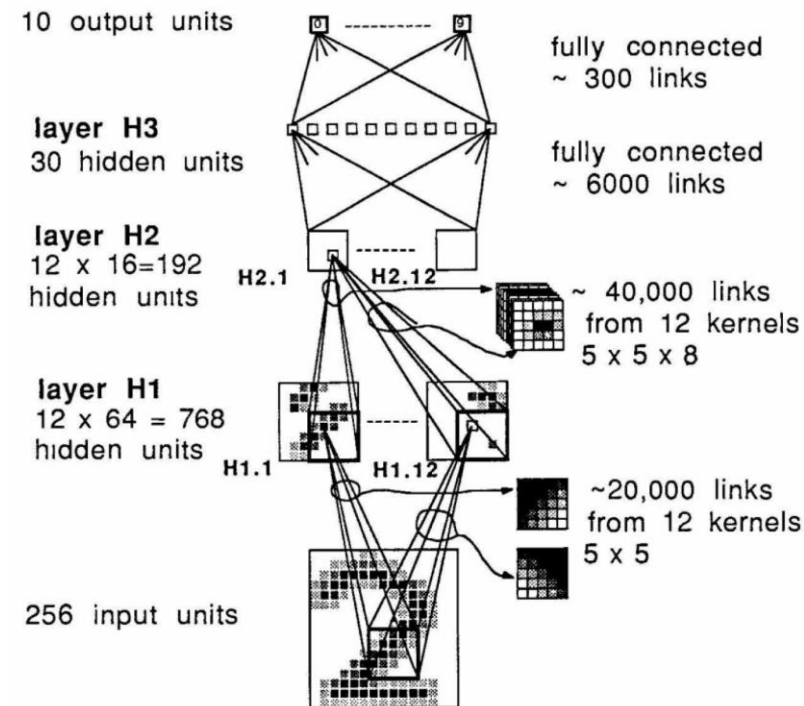
Alex Krizhevsky, Ilya Sutskever, Geoffrey E Hinton, 2012



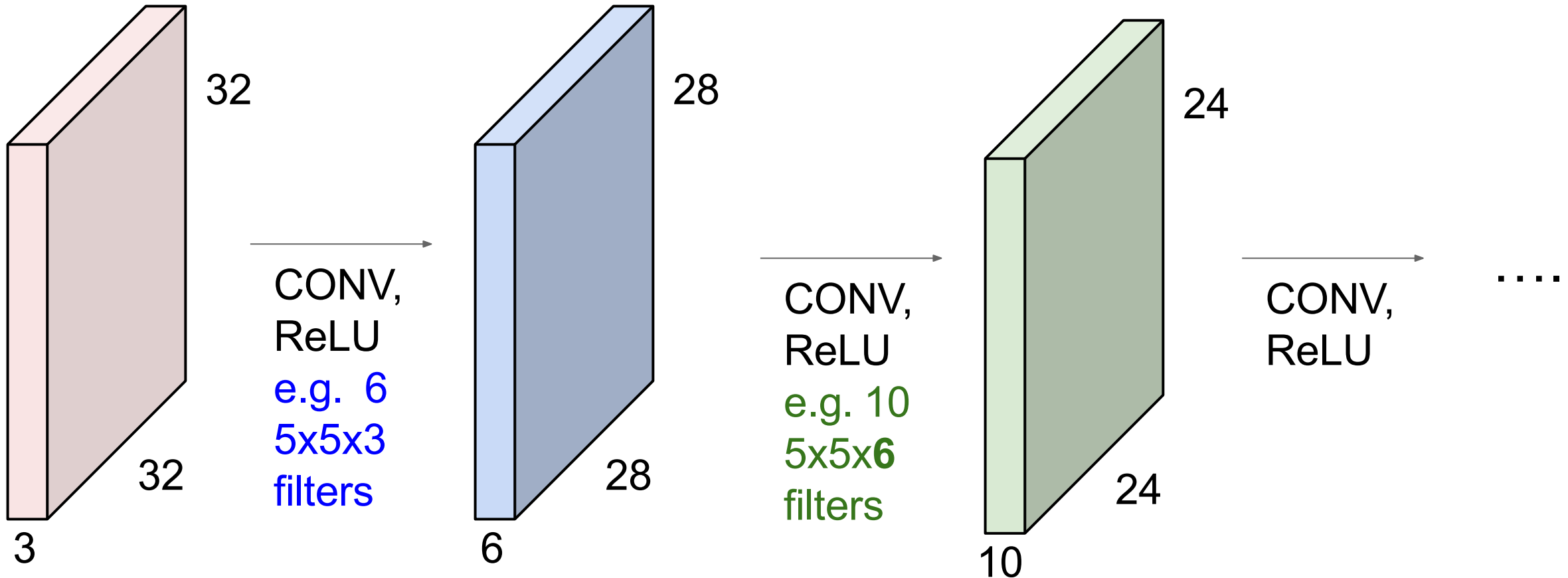
Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012.

Enabling Progress in DNNs

- Progress in modelling
 - ▶ Cognitron/Neocognitron [Fukushima 1971-1982]
 - ▶ Pooling [Riesenhuber and Poggio 1999]
 - ▶ Convnet's [LeCun et al. 1989]
 - ▶ Non-linearities [Nair, Hinton 2010]
 - ▶ DropOut [Krizhevsky et al. 2012]
 - ▶ Batch Normalization [Ioffe Szegedy 2015]
 - ▶ Identity mapping [He et al. 2015]
 - ▶ Attention [Bengio et al. 2015]
 - ▶ ...



ConvNet is a sequence of Convolution Layers, interspersed with activation functions



Examples time:

- Consider stride 1 and pad 1. What's the convolution output?

$$\begin{array}{c} g[k, l] \\ \begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline \end{array} \end{array} \otimes \begin{array}{c} I[k, l] \\ \begin{array}{|c|c|c|} \hline 8 & 5 & 2 \\ \hline 7 & 5 & 3 \\ \hline 9 & 4 & 1 \\ \hline \end{array} \end{array} = \begin{array}{c} f[m, n] \end{array}$$

Recall:

$$f[m, n] = I \otimes g = \sum_{k, l} I[m - k, n - l] g[k, l]$$

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Recall:

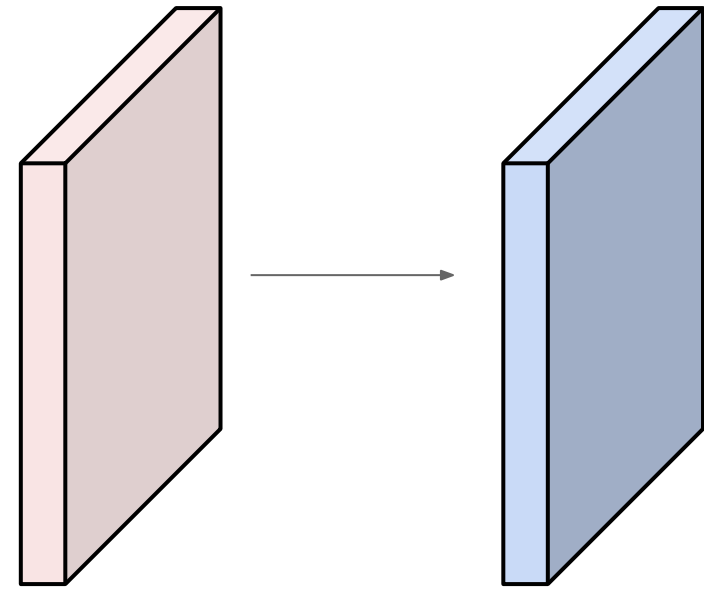
$$f[m, n] = I \otimes g = \sum_{k, l} I[m - k, n - l] g[k, l]$$

Examples time:

Input volume: **32x32x3**

10 5x5 filters with stride 1, pad 2

Output volume size: ?



Examples time:

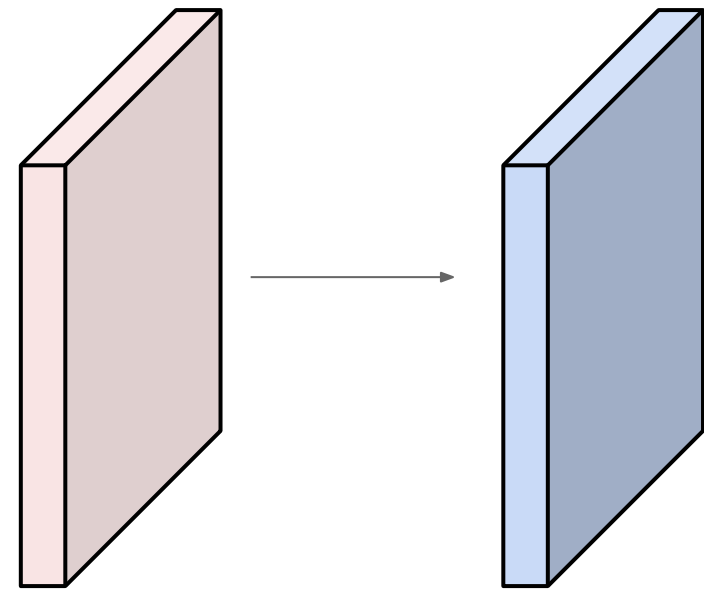
Input volume: **32x32x3**

10 **5x5** filters with stride **1**, pad **2**

Output volume size:

$(32 + 2 * 2 - 5) / 1 + 1 = 32$ spatially, so

32x32x10

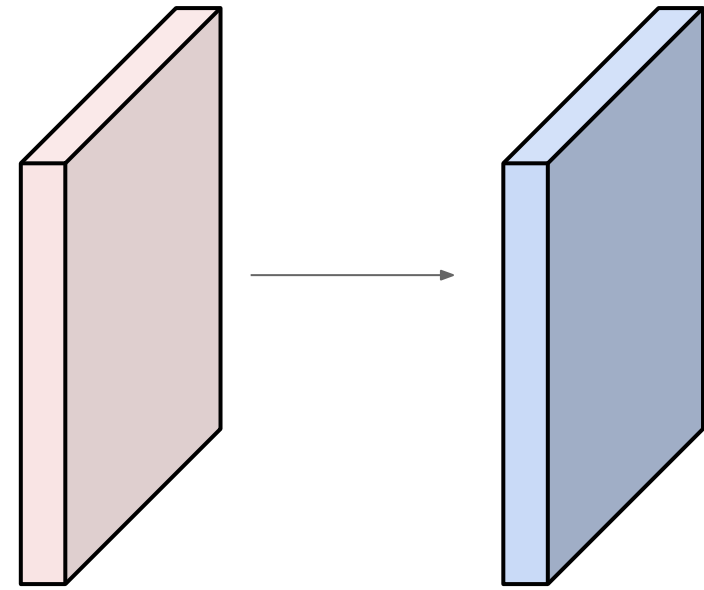


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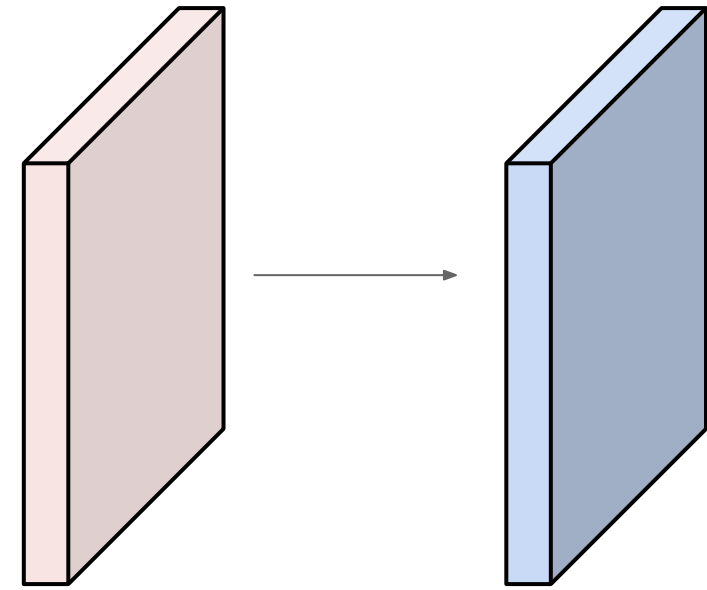
Number of parameters in this layer?



Examples time:

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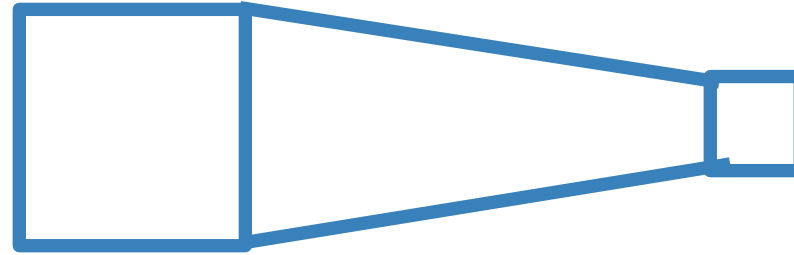
Number of parameters in this layer?

each filter has $5*5*3 + 1 = 76$ params (+1 for bias)

=> $76*10 = 760$

Receptive Fields

For convolution with kernel size K , each element in the output depends on a $K \times K$ **receptive field** in the input



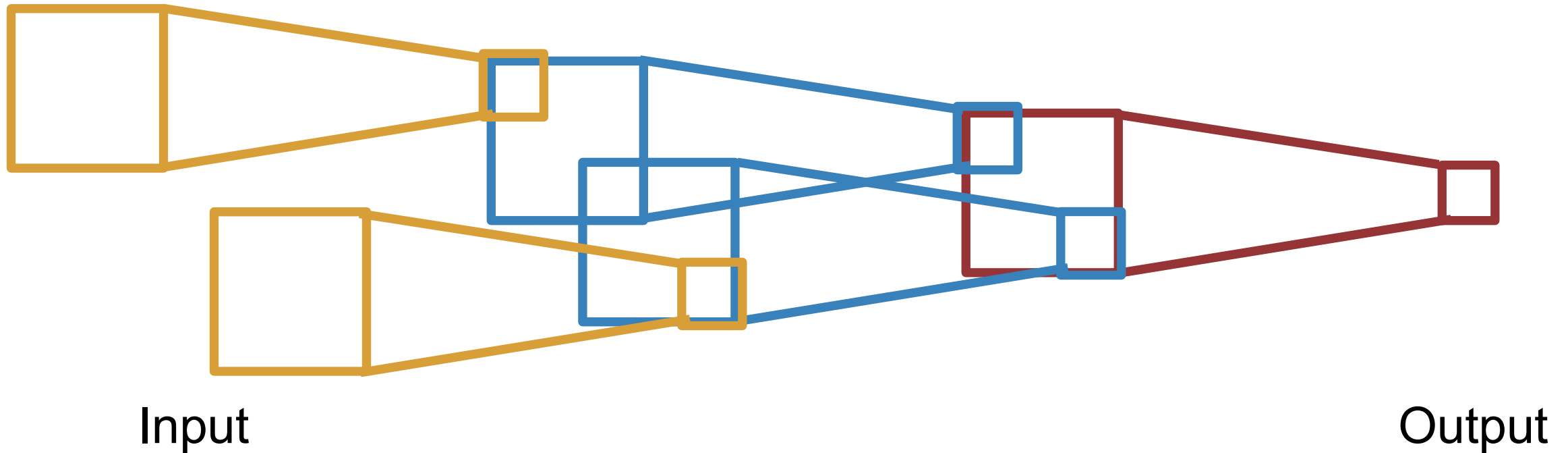
Input

Output

Slide inspiration: Justin Johnson

Receptive Fields

Each successive convolution adds $K - 1$ to the receptive field size
With L layers the receptive field size is $1 + L * (K - 1)$



Be careful – “receptive field in the input” vs. “receptive field in the previous layer”

Slide inspiration: Justin Johnson

Large Convnets for Image classification

- Convnets stack layers of convolution, non-linearity, pooling layers
- Trend towards smaller filters and deeper architectures
 - Smaller filters tend to be easier to learn than large ones

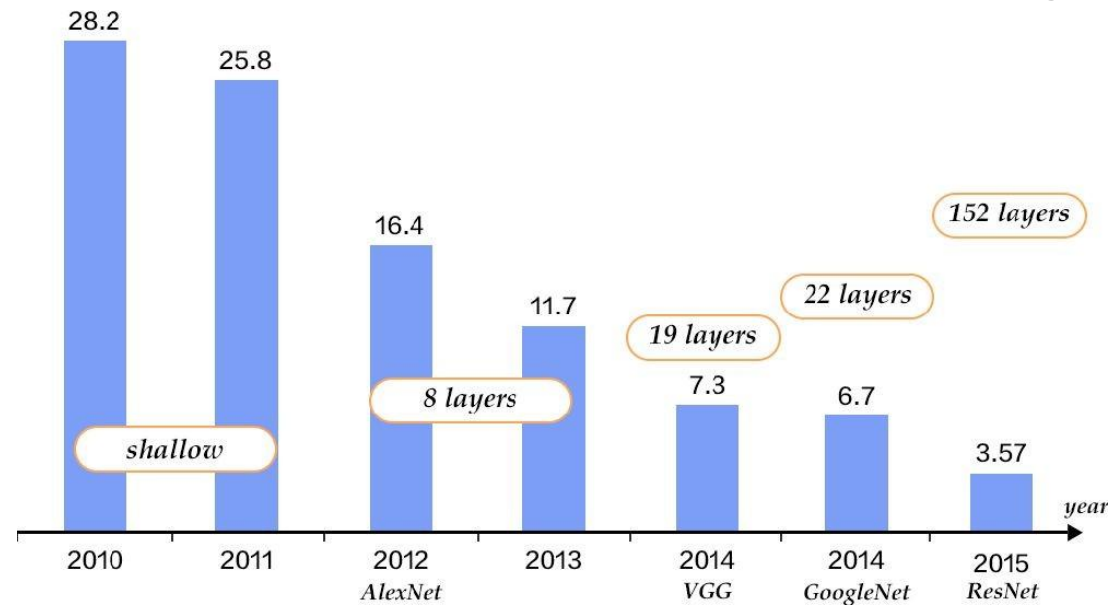


Figure Credit: Felsberg, Michael. "Five years after the Deep Learning revolution of computer vision : State of the art methods for online image and video analysis." (2017).

Additional reading

- Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, Dhruv Batra (2019). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. In IJCV'19 – also previously in ICCV'17
- Aditya Chattopadhyay, Anirban Sarkar, Prantik Howlader, Vineeth N Balasubramanian (2018). Grad-CAM++: Improved Visual Explanations for Deep Convolutional Networks. In Proc. WACV'18
- Petsiuk, V., Das, A., and Saenko, K. (2018). Rise: Randomized input sampling for explanation of black-box models. In Proc. BMVC'18
- Bach, S., Binder, A., Montavon, G., Klauschen, F., Mueller, K.-R., and Samek, W. (2015) LRP: On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. In PloS one'15
- Alaa Anani, Tobias Lorenz, Mario Fritz, Bernt Schiele (2025). Pixel-level Certified Explanations via Randomized Smoothing. ArXiv preprint.

Thank you

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